

RISK AVERSION AND MONETARY POLICY – EVIDENCE FROM INDIAN ECONOMY USING VAR MODEL

Sonalika Sinha¹

July, 31st, 2015

ABSTRACT

This paper examines how monetary policy influences risk taking and uncertainty in the Indian economy. Examined in the domestic context, the study shows a close relationship between the policy rate chosen by the Reserve Bank of India (RBI) and measured risks as given by the INDIA VIX index of implied volatility on Indian equity options. This paper confirms existence of a strong co-movement between INDIA VIX i.e. the expected stock market options based volatility, and the monetary policy stance. The VIX index is decomposed into two components – Risk Aversion and expected stock market volatility (Uncertainty). Results reveal that a lax monetary policy has a stronger impact on decreasing risk aversion, as compared to decreasing uncertainty. Results hold in a structural autoregressive model, controlling for multiple identification schemes and business cycle fluctuations.

JEL CODES: E44, E52, G12, G20

1. INTRODUCTION

This paper examines how monetary policy influences risk taking and uncertainty in an economy. Examined in the domestic context, the study shows a close relationship between the policy rate chosen by the Reserve Bank of India (RBI) and measured risks as given by the INDIA VIX index of implied volatility on Indian equity options.

The findings of the paper confirm that a fall in RBI's policy rate is followed by a dampening of the INDIA VIX index. Specifically, the study reveals that a loosening or lax monetary policy reduced volatility risk premium in stocks after about 12 months with a persistent effect lasting for more than three years. This paper confirms existence of risk aversion measures being

¹ Research Scholar, University of Hyderabad, School of Economics, sonalika.snh@gmail.com

closely tied to monetary policy actions. The INDIA VIX, therefore, proves to be a crucial link between monetary policy, risk aversion and uncertainty.

Recent literature witnesses growing concern across the globe of risks involved in persistently accommodative monetary policy tools. This issue particularly shot into prominence in the aftermath of the global financial crisis in 2008. The period that led up to the crisis, popularly known as the Greenspan put was characterized with highly accommodative monetary policy that led to decreasing risk aversion among investors, attributed as one of the crucial reasons for the breakout of the global financial crisis. Though several studies were conducted to analyze risk, uncertainty and financial crises prior to 2008, Borio and Zhu (2012) established the 'risk-taking channel' of monetary policy. Low interest levels for a continued period of time increase the risk-taking tendencies among banks and financial investors. Rajan (2005) discussed the increased levels of risk-taking in the economy owing to excessive liquidity supplied by central banks. This paper confirms similar evidence of decreased risk-aversion in times of accommodative monetary policy in India.

The analysis requires four components to be analyzed thoroughly – risk aversion, uncertainty, monetary policy stance as well as a business cycle indicator. The data used in the analysis spans from March 2009 to June 2015. The paper employs daily data of 2300 observations through 2009 to 2015. The sources for monetary policy data and business cycle indicator is from the RBI. The VIX and CNX NIFTY returns data is acquired from National Stock Exchange (NSE). Though the entire data is available from earlier periods, this paper focuses on the financial (in)stability caused in the post crisis environment due to lax monetary policies sustained over a period of time. The VIX daily data is available from November 2007 by CBOE, however this paper deliberately excludes the crisis period for the sake of stability in the data.

The paper confirms existence of strong co-movement between VIX and measure of MP in the short run in India. The VIX is a popularly known measure of risk aversion in financial markets. Results show an existence of a strong co-movement between INDIA VIX i.e. the expected stock market returns, and the monetary policy stance. The paper considers the cross correlogram between the real interest rate, as the monetary policy stance and the logarithm of end-of-month readings of the INDIA VIX index. The VIX index essentially measures the risk-neutral expected stock market variance for the CNX NIFTY 50 index.

-Figure 1 here-

The correlogram reveals a strong co-movement between log of VIX and the monetary policy stance. The purpose of analyzing the interaction between VIX and monetary policy has a threefold objective. First, the paper examines implications of strong co-movement of VIX, also known as the ‘fear index’, with monetary policy. This strong interaction has become of particular significance in the post crisis financial literature which rekindled literature wherein loose monetary policy may lead to increased and excessive risk taking in financial markets.

Second, the monetary policy is known to respond to economic conditions prevalent in the economy. Particularly, monetary policy may be conceived to be responsive to the uncertainty shocks in the economy, so as to impact economic outcomes. The VIX index is often used to measure uncertainty as used by Bloom (2009). However, the VIX index can be decomposed into two components – RA and UC. To explain, the VIX indicates actual expected stock market volatility which is seen as a measure of uncertainty. However, the decomposition also yields a residual of the VIX, popularly known in financial literature as the Variance Premium. This variance premium is calculated as a measure of risk aversion. This paper examines which aspect of the VIX, after decomposition, is found to drive the strong co-movement of VIX and the MP stance. Third, this study examines the relationship between monetary policy and the stock market in India.

In a first of such an attempt in India, this paper establishes evidence of link between risk aversion and monetary policy implications. The VIX is characterized as the risk neutral expected stock market variance or a measure of implied volatility. The analysis carried out by Bekaert and Hoerova (2013) reveals the conditional variance of stock market to be a better indicator of financial instability than the variance premium. In order to proxy the VIX as an effective measure of economic activity and financial instability, this paper decomposes the squared VIX index into conditional stock market variance and equity variance premium.

2. LITERATURE REVIEW

This paper uses the “fear index” measured among financial investors to calculate the risk-aversion level during expansionary monetary policy. In a similar attempt, Bekaert and Hoerova (2013) take the step to decompose the VIX index to analyze predictive powers of the index for stock market returns and economic activity. The VIX index is chosen as the volatility index

measure to account for the risk-taking capacity in financial markets. It is first crucial to establish whether the tool adopted to measure risk aversion is suitable. Thenmozhi and Chandra (2013) establish INDIA VIX is a fear index that accurately captures risk aversion among investors. In addition, they also prove INDIA VIX to be a far better instrument to analyze volatility in comparison with traditional measures of volatility. However, in the context of analyzing risk aversion among investors in general, specifically not limited to the scope of analyzing stock market returns, it is crucial that the VIX index has a contemporaneous correlation with financial instability in the economy. In an analysis for the United States, Bekaert, Hoerova, Lo Duca (2012) show strong interactions between monetary policy and variance premium (calculated from the VIX index). This adds further to the suitability of the using VIX measure for determining risk aversion. To make the VIX index appropriate for a measure of risk-aversion, Bekaert and Hoerova (2013) decompose the squared VIX index derived from US S&P500 options prices in order to examine the predictive power of VIX and its components for indicating economic activity. They find conditional stock market variance predict economic activity. This paper effectively proves that VIX is an appropriate for using VIX as risk-aversion proxy that accounts for overall economic level activity in India.

3. MEASUREMENT

This paper establishes a link between monetary policy and risk-aversion among investors, thereby empirically confirming the evidence of risk-taking channel of monetary policy in India. Using a VAR framework approach to modelling the link between monetary policy stance and risk aversion tendencies among investors, the findings posit a strong co-movement of the VIX index with monetary policy. Further, the co-movement gets stronger after a lag period. The model answers questions about the persistence of risk-channel effects and also controls for changes in business cycle movements.

3.1 DATA

The data for daily INDIA VIX and Realized Volatility are derived from CMIE Prowess and National Stock Exchange (NSE, Historical data since 2009). Other variables are obtained from the Handbook of Statistics, Reserve Bank of India. This paper utilizes daily data for the period

after the global financial crisis i.e. through 2009-2015. As the CBOE-approved VIX data started for Indian markets since March 2009, the paper analyzes model-free INDIA VIX data since 2009.

The VIX index used by US S&P500 was replicated for Indian financial markets in 2009. While the CBOE was the first exchange to establish such a volatility index in 1993, the Indian volatility index (henceforth, INDIA VIX) was introduced in 2009 as a measure of implied volatility on the CNX NIFTY50 index. The methodology used for determining INDIA VIX is identical to the one adopted by CBOE for S&P500 options prices, thereby giving reason to empirically test for risk aversion amidst monetary policy effects using INDIA VIX as a good measure.

3.2 MEASURING RISK AVERSION AND UNCERTAINTY

This paper follows the model proposed by Bekaert et al (2013), the VIX index is decomposed into two components – risk aversion and uncertainty. INDIAVIX as it is termed in Indian markets, is a volatility measure based on the NIFTY Index Option prices – indicating market volatility over the next 30 business days in the market. The calculation method used for measuring INDIAVIX is based on the method used by the Chicago Board of Options Exchange (CBOE). The index first started in India since 2009 and this paper uses a model-free index of implied volatility to capture the impact of monetary policy and the link with risk-aversion.

The empirical exercise begins with decomposing the INDIAVIX index into two components – Risk aversion and Uncertainty. INDIA VIX is a volatility measure based on the CNX NIFTY Index Option prices – indicating market volatility over the next 30 business days in the market. The calculation method used for measuring INDIAVIX is based on the method used by the Chicago Board of Options Exchange (CBOE). This paper uses a model-free index of implied volatility to capture the impact of monetary policy and the link with risk-aversion using daily data of INDIA VIX starting March, 2009.

In order to decompose the VIX index into two measures, there is a need to estimate the expected future realized variance. The decomposition is performed by projecting future realized volatility (hereafter RVAR) onto a set of current instruments. This approach requires daily data on monthly realized variances (RVAR, calculated using 22 day trading window), the squared

VIX and dividend yield. Daily data is used for this approach to gain better statistical power compared to other studies that follow end-of-month data.

RVAR is calculated using intraday returns on the CNX NIFTY 50 index. The realized variance is calculated by the formula:

$$RVAR=[252/n * \sum R_t^2]*100$$

where,

R_t : Daily intraday returns of NIFTY index and 'n' is the number of trading days taken into consideration. As this paper requires monthly realized variance, the number of trading days are taken as 22.

The next step is to select a good forecasting model. For this purpose, four volatility forecasting models are made to compete. Four OLS models use OLS regressions with predictors: a one-variable model with either past realized variance or the squared VIX; a two-variable model with both squared VIX and past RVAR; and a three-variable model adding the past dividend yield². The model selection criteria out-of-sample root mean squared error and mean absolute error values. Both in-sample and out-of-sample OLS forecasting is used in a recursive estimation. The regressions are also compared for winsorized and non-winsorized samples. The procedure leads to the two-variable model emerging as the best predictor. The selected model is the most stable forecasting model with economically significant coefficients. Calculation of Realized Volatility:

In financial literature, volatility is broadly classified into three categories – i) Physical ii) Historical/ realized volatility and iii) Implied Volatility. It is the second component that is calculated using CNZ NIFTY returns data. There are other papers that calculate this RVOL using 5 minute returns, however for the scope of this paper, it was deemed sufficient to capture RVOL using daily return as the VIX index is also an individual index. Further regressions explain how VIX and RVAR are fitted together to attain measures of the desired decompositions.

² Dividend yield on the CNX NIFTY 50 index, daily

4. VAR ANALYSIS

The SVAR approach is adopted as it allows imposing restrictions on the contemporaneous and lagged matrices of coefficients to improve estimation results. While the impulse responses are of particular interest, the reduced form VAR is estimated first. The VAR includes 4 lags, as chosen by the Akaike and LR criteria. The AIC minimises the forecast error variance.

The analysis follows the identified monetary VAR literature to set up a benchmark VAR that contains four variables - i) Risk aversion (RA); ii) Uncertainty (UC); iii) Monetary policy rate used by the RBI (MP); and, iv) Business cycle proxy (BC). The first two components are proxies using the VIX index decomposition techniques explained in earlier sections. The real rate of interest is considered as a measure of monetary policy stance and the business cycle indicator is represented using log difference of the IIP index. The log difference of IIP is the most crucial control variable in the VAR system since it is imaginable that RBI makes changes in their monetary policy stance owing to lesser than expected growth in the economy that may simultaneously lead the central bank to make cuts in their policy rate. Such a phenomenon is likely to render investors more risk averse in fear of an uncertain future.

The four variables may be represented in the form of a vector Z_t :

$$Z_t = [BC_t, MP_t, RA_t, UC_t]$$

Ignoring the constants, the structural VAR may be represented as:

$$A Z_t = \Phi Z_{t-1} + \varepsilon_t$$

Where A is a 4×4 full-rank and $E[\varepsilon_t \varepsilon_t'] = I$.

The selection of the number of variables considers Sims's methodology (1980) in selecting variables and their lag length. The variables included in the VAR are selected according to the economic interpretation. In this case, the variables are selected following the interaction of monetary policy and the decomposed components of the VIX index.

The impact of shocks is identified by writing a vector autoregression in recursive form, following Sims (1980) wherein we impose restrictions on the primitive system. In the four-variable VAR, the Cholesky restrictions result in the following exclusion restrictions on contemporaneous responses in the matrix A to fit a just-identified model.

The ordering of the variables imposed in the recursive form implies that the variable with index 1 is not affected by the contemporaneous shocks to the other variables, while variable 2 is affected by the contemporaneous shock to variable 1, but not variables 3 and 4.

As a result, variables that have slower impact, like IIP, are better candidates to ordered earlier than fast moving variables like RA and UC. In this paper, the business cycle indicator is therefore ordered first, followed by monetary policy indicator i.e. the real interest rate, followed by risk aversion and uncertainty is ordered last. According to the Cholesky decomposition of the estimate of the variance covariance matrix, a total of six restrictions on the VAR are needed to identify the system. These six restrictions placed follow from the fact that in a n-variable VAR, as there are n regression residuals and n structural shocks, exact identification requires that $(n^2-n)/2$ restrictions be placed. As a result, Cholesky decomposition of the matrix A makes it a lower triangular system.

The next set of restrictions follows from what is suggested by economic models – that money has no long run effect on industrial production. This long run literature stems from long run money neutrality that money should not have any long run effect on real variables. Such a theory inspires a Blanchard and Quah (1989) decomposition. The Blanchard and Quah methodology imposes long run restrictions on the impulse response functions to exactly identify the structural innovations. This feature provides a unique decomposition of an economic time series into its temporary and permanent components. The bootstrapped 90% confidence intervals are based on 1000 replications.

4.1 Impulse response graphs:

Figure 2 presents all the impulse response functions from the four-variable recursive VAR with 90 percent confidence bands. Represented in key panel form, the graphs provide the main findings of the narrative.

-Figure 2-

Consider first the impact of a shock to the real interest rate, and study the impact of this shock to risk aversion. Panels A and B show the interaction between the real interest rate (RERA) and risk aversion (RA).

A one standard deviation negative shock to the real rate decreases risk aversion under both contemporaneous and long run contemporaneous identifications. Low monetary policy lowers risk aversion for the first six months, reaching a maximum of 0.006 in both models. The impact

reverses immediately to increase risk aversion to reach a minimum at over a one year period. The impact remains significant for 6 years. A one standard deviation shock to risk aversion has a mostly positive impact on real interest rate, however it remains statistically insignificant throughout the period.

In Panel C, we find that a positive shock to monetary policy has positive impact on uncertainty. However, this impact dies quickly over 6 months and reverses in the next 6 months reducing uncertainty to a minimum over 24 months. Over the medium-run, the impact is positive on uncertainty. In Panel D, we observe the real rate decreases in the short-run following a one standard deviation positive shock to uncertainty. However, the impact is statistically insignificant under both identification schemes.

Panel E reveals that expansionary monetary shock has a positive impact on IIP for the first 6 months, and the impact turns negative after a lag of 24 months as the monetary shock becomes expansionary. In Panel F monetary policy reacts as expected to business cycle fluctuations, a one standard deviation positive shock to IIP leads to a higher real rate. The interactions between risk aversion on IIP remains statistically insignificant. Positive shocks to uncertainty lower industrial production growth after a lag of 18-24 months. The responses remain statistically significant. This implies that using VIX as a measure of uncertainty reveals how shocks in uncertainty generate significant business cycle effects.

Finally, Panel K shows how increases in risk aversion predict future increases in uncertainty under both contemporaneous and long run contemporaneous identifications. Uncertainty, as shown in Panel K has a positive impact on risk aversion after a lag of 9-12 months.

CONCLUSION

This paper examines which component of the VIX drives its strong co-movement with the monetary policy stance in the short run – Uncertainty or Risk Aversion. In other words, the question that arises is whether the strong co-movement between monetary policy and VIX index is driven more by. The VIX index is decomposed into two components – Risk Aversion measure (hereafter RA) and Uncertainty measure (hereafter UC). Such analysis helps better understand the relationship between monetary policy and the stock market, which has been examined in depth in the extant literature.

This paper employs a VAR model to empirically establish a link between Indian monetary policy rates, given by the Reserve Bank of India (RBI), and its impact on risk-taking or risk-averse tendencies in the economy. The model also includes a control for business cycle components, based on the model proposed by Bekaert et al (2013), to account for business cycle variations and rule out any joint effects of business cycle changes on market volatility. The level of risk may be associated with the stage of the business cycle or recession in the economy, which is reflected by the monetary policy. Therefore, in order to single out the effects of the monetary policy on risk-taking channels, the model proposed controls for business cycle movements using a structural vector autoregressive framework.

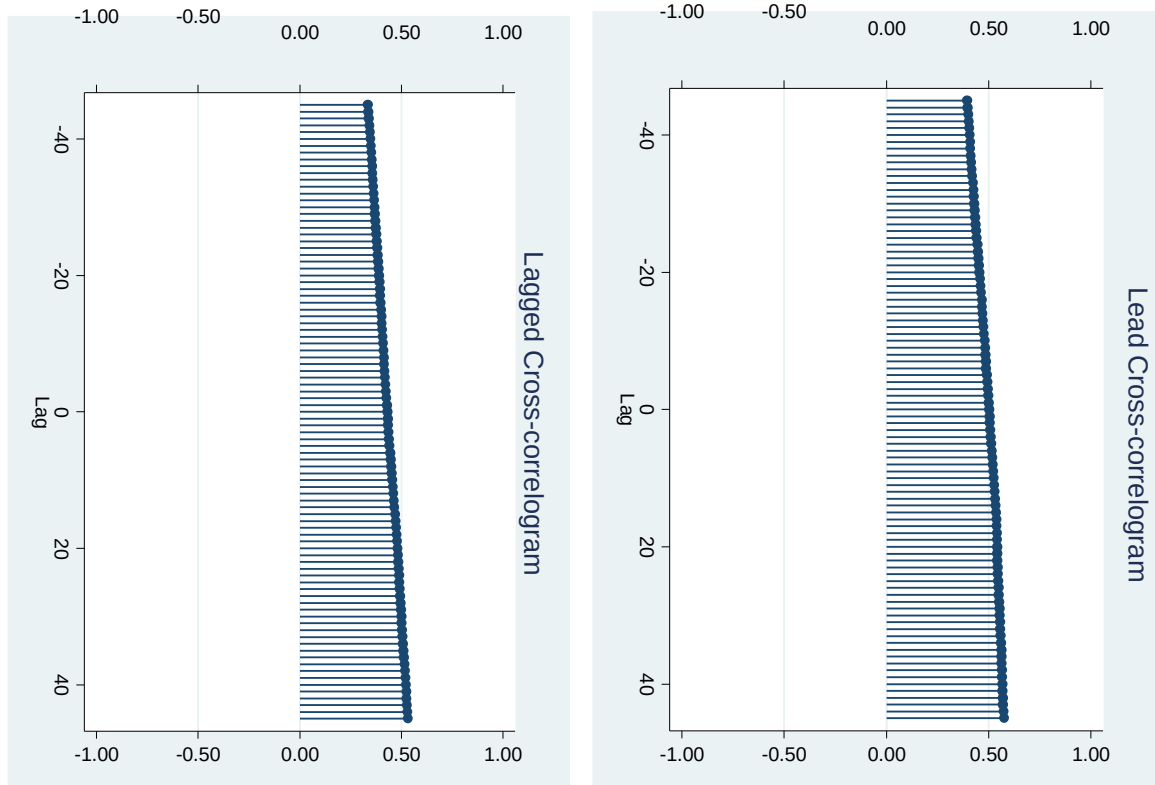
The main findings of plotting IRFs confirm that monetary policy has a medium-run statistically significant effect on risk aversion. This effect is also economically significant. The findings of the paper confirm that a fall in RBI's policy rate is followed by a dampening of the INDIA VIX index. Specifically, the study reveals that a loosening or lax monetary policy reduced volatility risk premium in stocks after about 12 months with a persistent effect lasting for more than three years. This paper confirms existence of risk aversion measures being closely tied to monetary policy actions.

REFERENCES

- Adrian, T., Shin, H. S., 2008. Liquidity, Monetary Policy, and Financial Cycles. *Current Issues in Economics and Finance* 14 (1), Federal Reserve Bank of New York.
- Altunbas, Y., Gambacorta, L., Marquéz-Ibañez, D., 2010. Does Monetary Policy Affect Bank Risk-taking? ECB Working Paper No. 1166.
- Bekaert, G., Hoerova, M., LoDuca, M., 2013. Risk, uncertainty and monetary policy. *Journal of Monetary Econ.* 60(7), 771–788.
- Bernanke, B., Kuttner, K.N., 2005. What Explains the Stock Market's Reaction to Federal Reserve Policy? *Journal of Finance* 60 (3), 1221-1257.
- Blanchard, O., Quah, D., 1989. The Dynamic Effects of Aggregate Demand and Supply Disturbances. *American Economic Review* 79 (4), 655-73.
- Borio, C., Zhu, H., 2008. Capital Regulation, Risk-Taking and Monetary Policy: A Missing Link in the Transmission Mechanism? BIS Working Paper No. 268.
- Chicago Board Options Exchange, 2004. VIX CBOE Volatility Index. White Paper.
- Rajan, R., 2006. Has Finance Made the World Riskier?" *European Financial Management* 12 (4), 499-533.
- Whaley, R.E., 2000. The Investor Fear Gauge. *Journal of Portfolio Management*, Spring, 12-17.

Figure 1

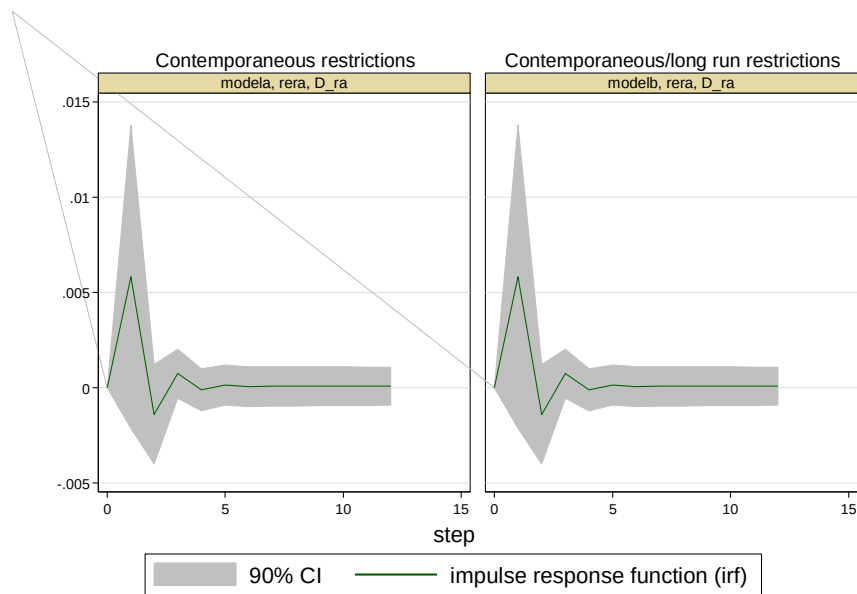
CROSS CORRELOGRAM LVIX RERA



Notes: The first column displays the lagged correlogram between the log of the INDIA VIX and i.e. LVIX and the past values of real interest rate (RERA). The second column displays the lead correlogram between the LVIX and future values of RERA. The confidence interval taken for the cross-correlation is 95%. The sample period is daily data between March 2009 and May 2015.

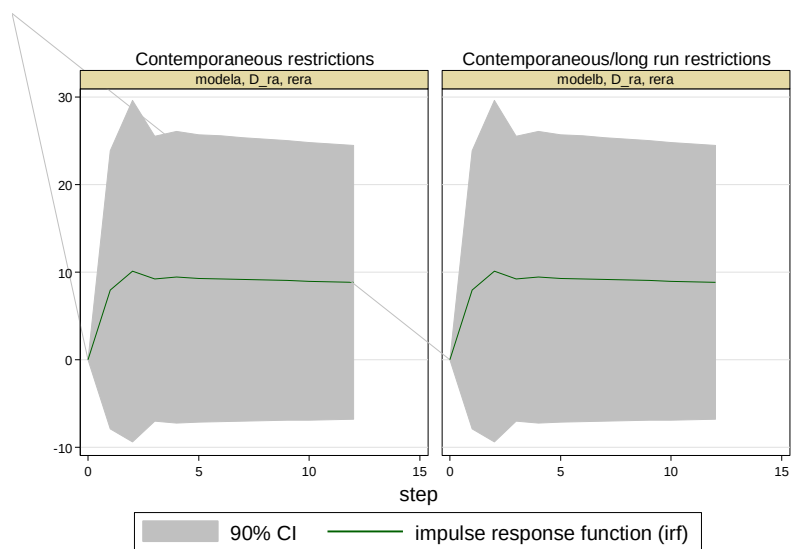
FIGURE 3: STRUCTURAL-FORM IRFS FOR THE FOUR-VARIABLE SVAR (IIP, RERA, RA, UC)

Panel A: Impulse RERA, Response RA



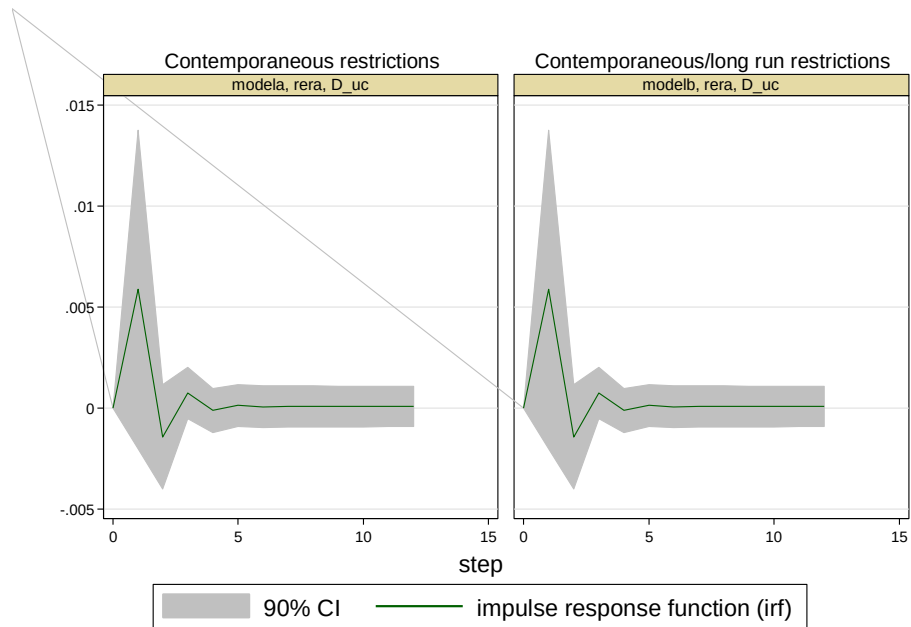
Graphs by irfname, impulse variable, and response variable

Panel B: Impulse RA, Response RERA



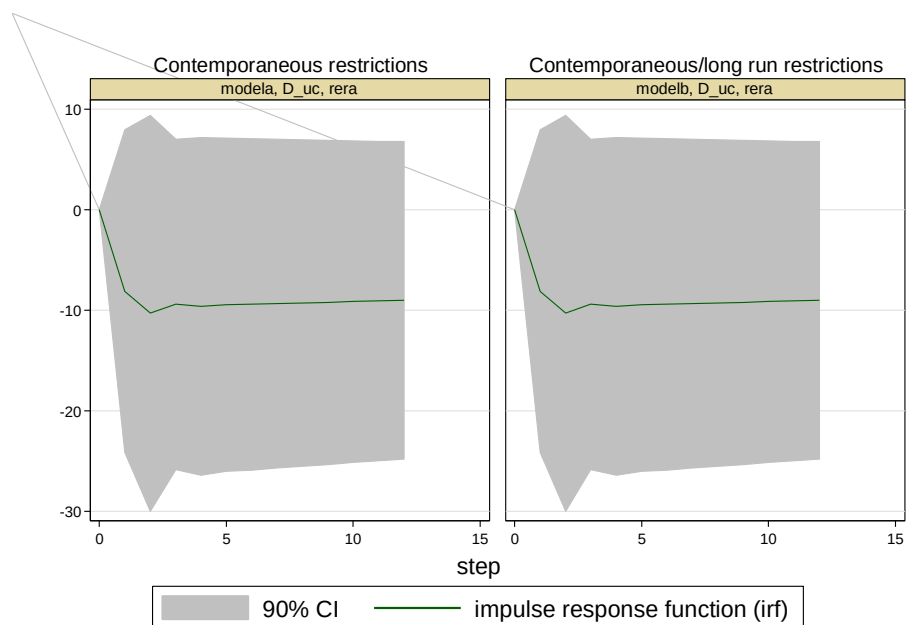
Graphs by irfname, impulse variable, and response variable

Panel C: Impulse RERA, Response UC



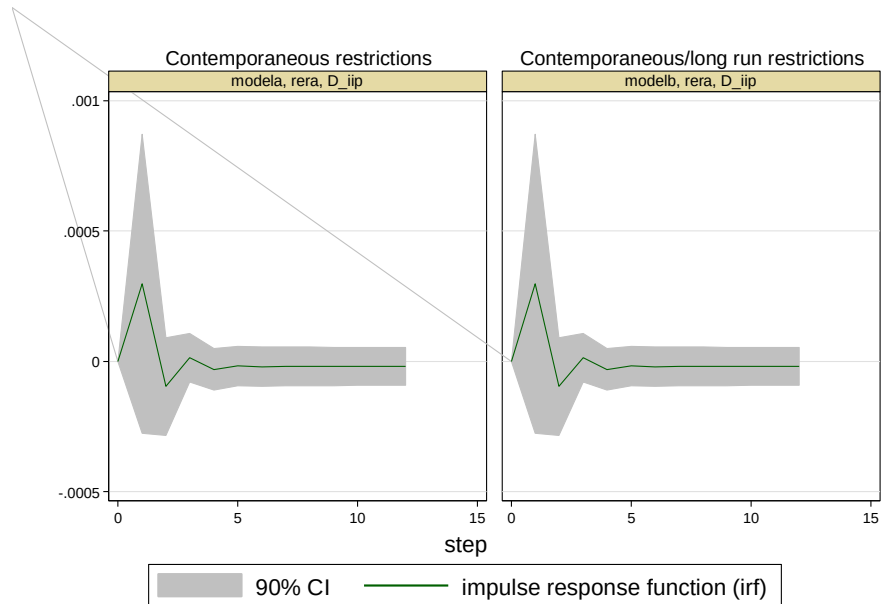
Graphs by irfname, impulse variable, and response variable

Panel D: Impulse UC, Response RERA



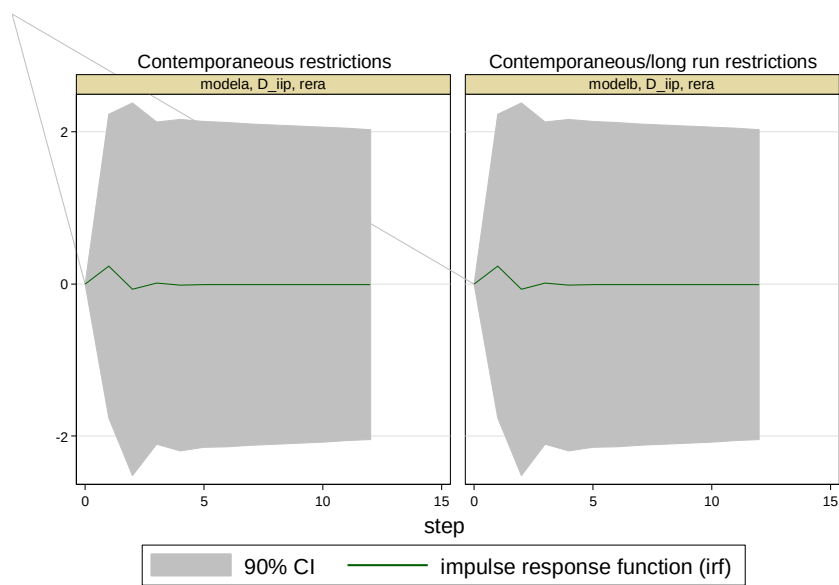
Graphs by irfname, impulse variable, and response variable

Panel E: Impulse RERA, Response IIP



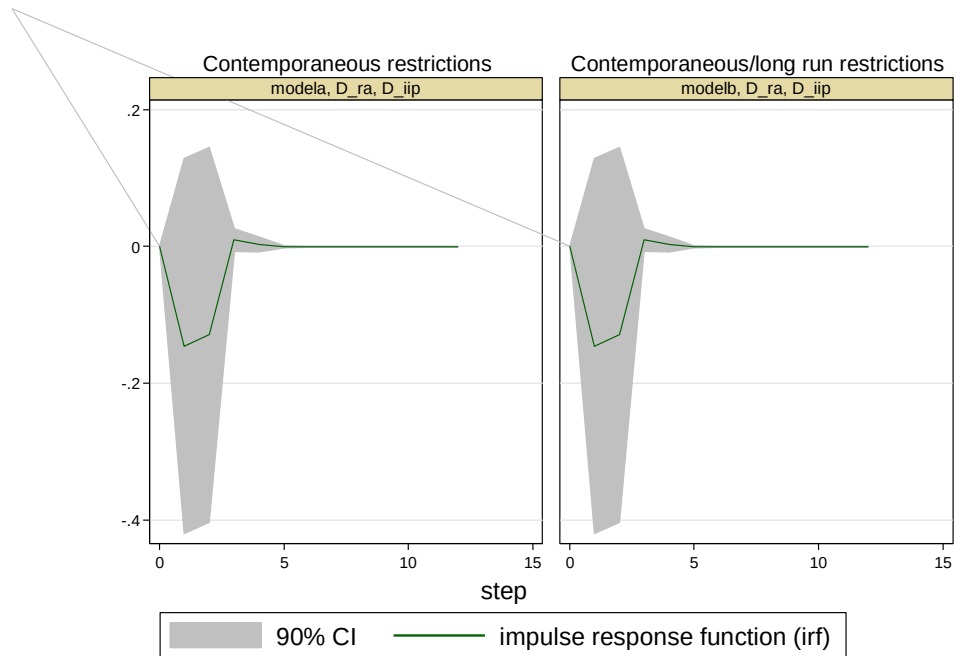
Graphs by irfname, impulse variable, and response variable

Panel F: Impulse IIP, Response RERA



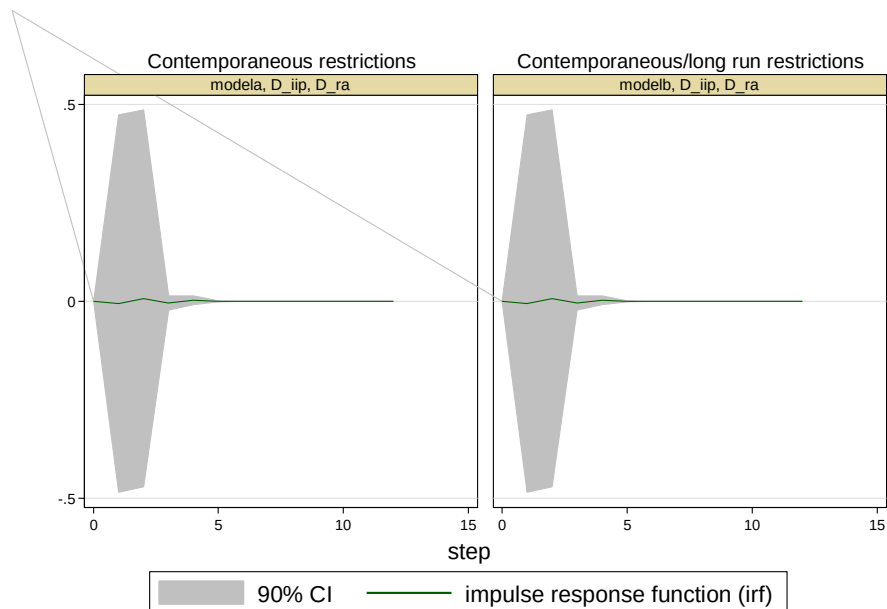
Graphs by irfname, impulse variable, and response variable

Panel G: Impulse RA, Response IIP



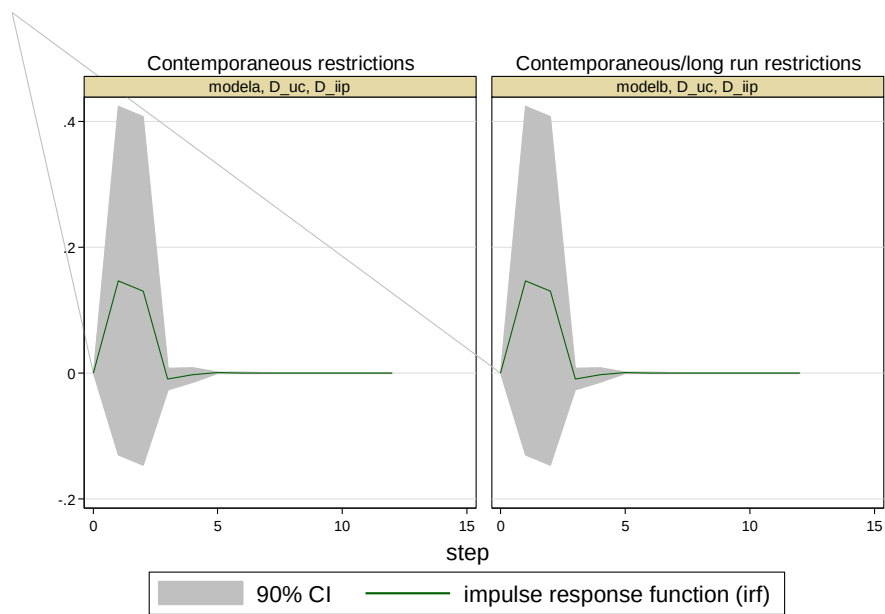
Graphs by irfname, impulse variable, and response variable

Panel H: Impulse IIP, Response RA



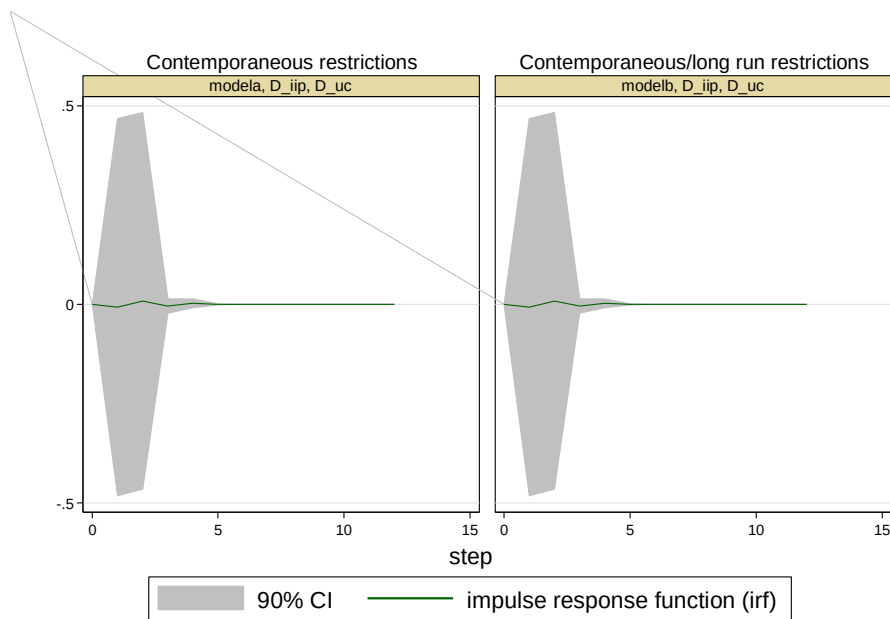
Graphs by irfname, impulse variable, and response variable

Panel I: Impulse UC, Response IIP



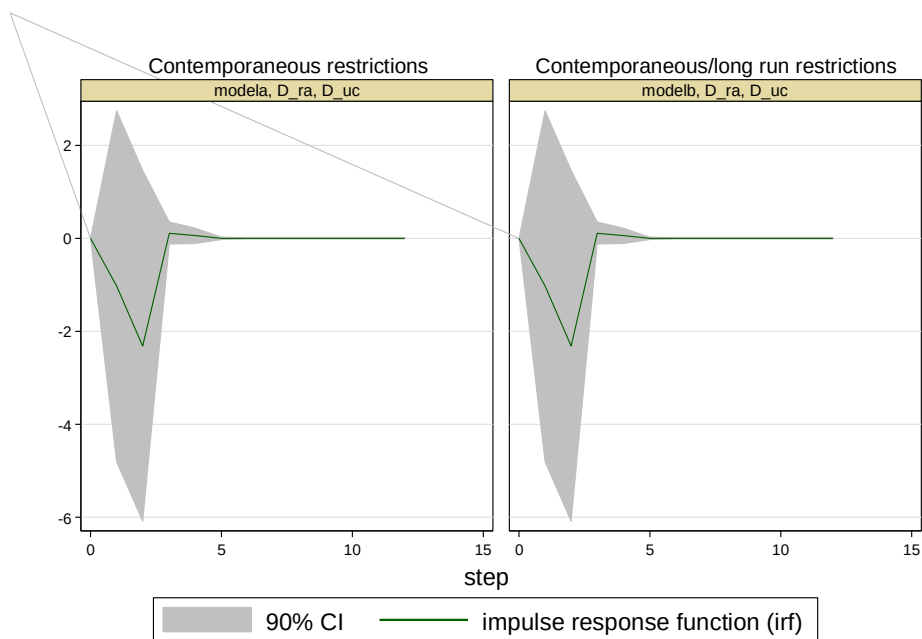
Graphs by irfname, impulse variable, and response variable

Panel J: Impulse IIP, Response UC



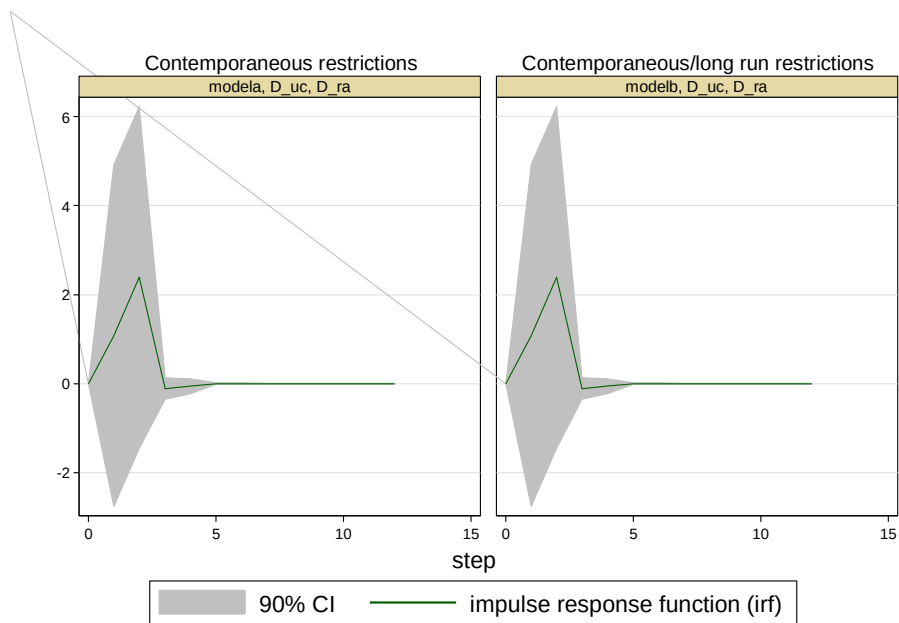
Graphs by irfname, impulse variable, and response variable

Panel K: Impulse RA, Response UC



Graphs by irfname, impulse variable, and response variable

Panel L: Impulse UC, Response RA



Graphs by irfname, impulse variable, and response variable